

DS3001 - A2

Q1 1) $m(a + bX) = a + b \cdot m(X)$

$$m(X) = \frac{1}{N} \sum_{i=1}^N X_i$$

$$m(a + bX) = \frac{1}{N} \sum_{i=1}^N (a + bX_i) = a + \frac{1}{N} \sum_{i=1}^N bX_i = a + b \cdot \frac{1}{N} \sum_{i=1}^N X_i = \boxed{a + b \cdot m(X)}$$

2) $\text{cov}(X, a + bY) = b \cdot \text{cov}(X, Y)$; $\text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (X_i - m(X))(Y_i - m(Y))$

$$\text{cov}(X, a + bY) = \frac{1}{N} \sum_{i=1}^N (X_i - m(X))((a + bY)_i - m(Y))$$

$$= \frac{1}{N} \sum_{i=1}^N (X_i - m(X))(a + bY_i - m(Y))$$

$$= \frac{1}{N} \sum_{i=1}^N (X_i - \frac{1}{N} \sum_{i=1}^N X_i)(a + bY_i - \frac{1}{N} \sum_{i=1}^N (a + bY_i))$$

$$= \frac{1}{N} \sum_{i=1}^N (X_i - \frac{1}{N} \sum_{i=1}^N X_i)(a + bY_i - (a + b \frac{1}{N} \sum_{i=1}^N Y_i))$$

$$= \frac{1}{N} \sum_{i=1}^N (X_i - m(X))(a + bY_i - a - b m(Y))$$

$$= \frac{1}{N} \sum_{i=1}^N (X_i - m(X))(bY_i - b m(Y))$$

$$= \frac{1}{N} \sum_{i=1}^N (X_i - m(X))(Y_i - m(Y))b$$

$$= b \cdot \frac{1}{N} \sum_{i=1}^N (X_i - m(X))(Y_i - m(Y))$$

$$\Downarrow$$

$$\boxed{= b \cdot \text{cov}(X, Y)}$$

$$m(X) = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y))$$

$$s^2 = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2$$

$$3) \text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X) \quad \& \quad \text{cov}(X, X) = s^2$$

$$\begin{aligned} \text{cov}(a+bX, a+bX) &= \frac{1}{N} \sum_{i=1}^N [(a+bX)_i - m(a+bX)][(a+bX)_i - m(a+bX)] \\ &= \frac{1}{N} \sum_{i=1}^N \left[a+bX_i - \frac{1}{N} \sum_{i=1}^N a+bX_i \right] \left[a+bX_i - \frac{1}{N} \sum_{i=1}^N a+bX_i \right] \\ &= \frac{1}{N} \sum_{i=1}^N [a+bX_i - (a+b \cdot m(X))][a+bX_i - (a+b \cdot m(X))] \\ &= \frac{1}{N} \sum_{i=1}^N b \cdot (X_i - m(X)) \cdot b (X_i - m(X)) \\ &= b^2 \cdot \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) \cdot (X_i - m(X)) \\ &= b^2 \cdot \text{cov}(X, X) \end{aligned}$$

$$\text{AND } = b^2 \cdot \frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2$$

$$\boxed{\text{cov}(a+bX, a+bX) = b^2 s^2}$$

$$\boxed{\text{cov}(X, X) = s^2}$$

$$4) m(g(X)) = g(m(X))$$

* $g()$ is non-decreasing $\Rightarrow g(x) = e^x$

$$m(g(X)) = \frac{1}{N} \sum_{i=1}^N g(x_i)$$

$$g(m(X)) = g\left(\frac{1}{N} \sum_{i=1}^N x_i\right)$$

$$g(x) = e^x$$

$$= \frac{1}{3} (g(1) + g(2) + g(3))$$

$$= g\left(\frac{1}{3} (1+2+3)\right)$$

$$= \frac{1}{3} (e^1 + e^2 + e^3)$$

$$= g\left(\frac{6}{3}\right) = g(2)$$

$$= \frac{e + e^2 + e^3}{3}$$

$$= e^2$$

$$\frac{e + e^2 + e^3}{3} \neq e^2 \Rightarrow \boxed{\text{False}}$$

$$g(x) = \ln x$$

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